# IPL Live Prediction

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CS1138 - Machine Learning

## Introduction & Motivation

- The Indian Premier League (IPL) is a fast-paced T20 cricket tournament with highly uncertain match outcomes.
- Predicting match results is challenging due to dynamic factors like player form, pitch conditions, and team strategies.
- Machine Learning enables data-driven prediction by analysing historical match data.
- Predict match outcome (Win / Loss)
- Identify key factors affecting match results

## Dataset & Key Insights

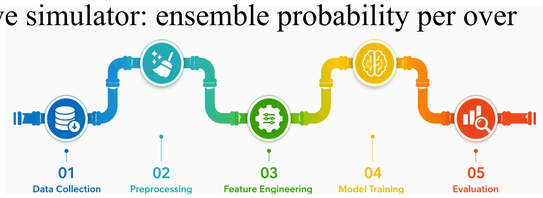
- 86K ball-by-ball records (IPL 2008–2025)
- 2nd innings only → target: batting_team_win
- Train: seasons < 2024 | Test: seasons ≥ 2024
- Live over-by-over win probability simulator

## Features

- Match State: runs_left, balls_left, wickets, crr, required_rr
- Engineered: pressure_index, run_rate_ratio, boundary_pressure, over_pressure, wickets×crr, balls_left²
- Contextual: home_advantage, toss_advantage, season
- Categorical: teams, venue, toss (One-Hot Encoded)

## Methodology & Feature Work

- Ball-by-ball → over-level match states (2nd innings)
- 6 cricket-specific interaction features engineered
- One-Hot Encoding for categorical variables
- Live simulator: ensemble probability per over




## Evaluation Metrics

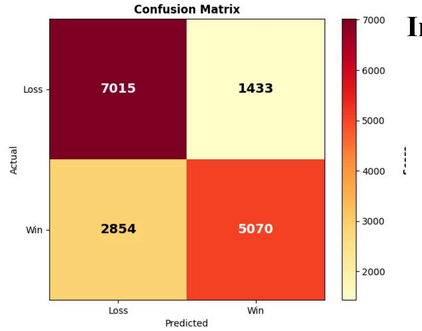
| Model | Accuracy | Precision | Recall | F1 Score | Log Loss |
|---|---|---|---|---|---|
| Logistic Regression | 73.82% | 0.78 | 0.64 | 0.7 | |
| Random Forest | 80.33% | 0.86 | 0.71 | 0.78 | 0.4462 |
| XG Boost | 80.38% | 0.86 | 0.71 | 0.78 | 0.4489 |

**Interpretation**

- Logistic Regression shows lower performance compared to ensemble models
- Random Forest and XGBoost achieve higher accuracy and better precision–recall balance
- XGBoost slightly outperforms Random Forest in overall accuracy

## Model I - Logistic Regression

- Baseline linear model with sigmoid probability output
- StandardScaler applied + class_weight='balanced'
- Strong at predicting Loss (Recall = 0.83) but weaker on Win class
- Serves as the lower-bound benchmark for ensemble models




**Interpretation**

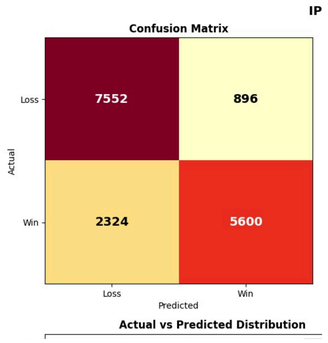
- Model predicts Loss better (Recall = 0.83) than Win
- Overall performance is moderate (~74%)
- Suitable as a baseline for comparison

## Model II - Random Forest

- Ensemble learning using multiple decision trees
- Handles complex and non-linear patterns
- Reduces overfitting compared to single models

**Interpretation**

- Achieves higher accuracy than Logistic Regression
- Shows better precision–recall balance
- Reduces misclassification errors significantly
- Strong model for IPL prediction

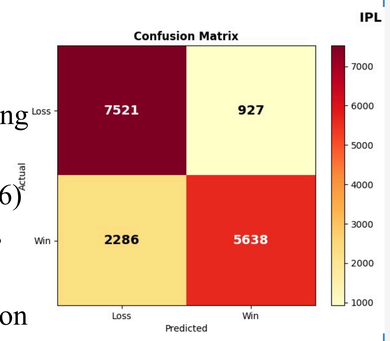



## Model III - XGBoost

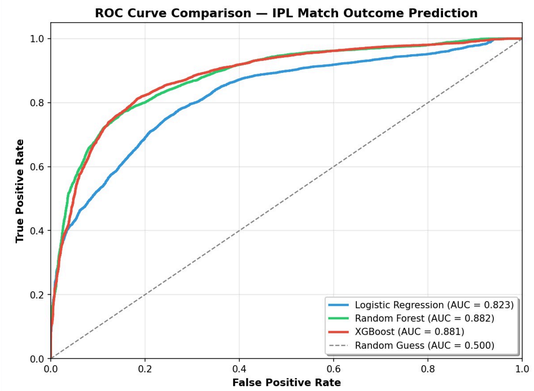
- Gradient boosting algorithm
- Builds models sequentially to reduce errors
- High performance on structured data

**Interpretation**

- Achieves the highest accuracy (80.38%), slightly outperforming Random Forest
- Shows balanced precision (0.86) and recall (0.71) across classes
- Overall, provides the most reliable and consistent prediction




## Results & Evaluation




**Model Comparison Summary**

- LR: 73.82% → linear baseline, limited by non-linearity
- RF: 80.33% → robust ensemble, captures interactions
- XGB: 80.38% → best overall (early stopping + boosting)
- Ensemble: Averages all 3 for live predictions

## Conclusion & Discussion

- ML models effectively predict IPL chase outcomes using live match state
- XGBoost (80.38%) marginally outperforms Random Forest; both strongly outperform LR
- Key insight: Match-state features (required_rr, pressure_index, boundary_pressure) dominate team/venue identity
- Live simulator enables real-time win probability tracking over-by-over with ensemble averaging
- Accuracy ceiling (~80%) reflects inherent T20 unpredictability (player form, weather, umpiring)